Where Cross-Question Structure in LLM Errors Comes From: Shared Seeds, Shared Prompts, and Answer-Letter Preference

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Abstract

When language models disagree on a question, the disagreement is often read as missing knowledge. If so, knowledge gaps should leave a trace: questions needing the same knowledge should be answered right or wrong together. We look for that trace beyond what chance or a model's overall ability explains. Within one model, where only the pipeline can couple questions, independent answers show none we can detect, but a trace appears when the questions share a random seed or few-shot examples. Across 24 models, the strongest trace beyond overall ability is a preference for one answer letter, mostly from one small model. When the options move to new letters, the trace follows the letter far more than the question. Counting a question correct only if it stays right in every rotation of the options removes this trace, at least where the preference acts on guesses. What little remains comes mostly from two pairs of near-duplicate models and partly from the small model. Why it remains is untested. In simulations, a second ability as strong as overall ability was usually detected, but a weaker one could be missed. Before reading disagreement as missing knowledge, sample independently and rotate the options.

1. Introduction

Self-consistency samples a language model K times on each question and reads disagreement among the samples as uncertainty about the answer (Wang et al., 2023; Manakul et al., 2023; Farquhar et al., 2024). In a deep ensemble (Lakshminarayanan et al., 2017), disagreement among members is read as epistemic uncertainty. Epistemic uncertainty comes from missing knowledge; aleatoric uncertainty is the noise that remains when the model is known (Kendall & Gal, 2017; Hüllermeier & Waegeman, 2021). Scores that add cross-model disagreement to self-consistency as an epistemic term (Hamidieh et al., 2026) make the same reading. Models' errors are related: they often choose the same wrong answer (Kim et al., 2025), and a few latent factors summarize many models' scores (Burnell et al., 2023; Ruan et al., 2024; Maimon et al., 2026). If disagreement reflects missing knowledge, a model that lacks some knowledge should fail together on the questions that need it. A per-question score cannot tell why models disagree, but the pattern across questions can test this prediction. We call the tendency of questions to be answered right or wrong together *cross-question structure*, and ask when it reflects shared knowledge and when it comes from something else.

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We call the kth samples of all questions *pass* k. Like one ensemble member, a pass answers every question once. Within one model, a knowledge gap lowers how often the passes answer a question correctly. It does not make two questions right or wrong together from pass to pass. For a fixed model, two questions can be coupled only by something the pipeline shares across them, such as the sampler's random numbers or the prompt. One seed in a batched call gives every prompt of a pass the same random numbers, and the standard 5-shot MMLU prompt places one set of exemplars before every question of a subject (Hendrycks et al., 2021). A within-model run is a control in which shared knowledge cannot create structure. Across models, ability couples the questions, and so can habits unrelated to knowledge. A model that prefers one answer letter is right more often on the questions whose correct option carries that letter (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024). Disagreement from missing knowledge should follow question content, and disagreement from a letter preference, the letter shown.

We measure cross-question structure by the eigenvalues of the question correlation matrix (App. A). Within a model, the reference is independent answers (Marčenko & Pastur, 1967; Johnstone, 2001). Across models, it is one ability per model, as in item response models (Rasch, 1960; Lord & Novick, 1968). A group of questions answered right or wrong together lifts an eigenvalue above its reference, and we call its eigenvector a *direction*. Within one model, runs that draw every sample independently show no direction, and a shared seed or prompt creates one (Sec. 3.1). Across

24 models the first direction is ability. The second is letter preference, which we identify by rotating the options to new letters (Sec. 3.2). We then count a question correct only if it is correct in every rotation, as in the CircularEval rule of Liu et al. (2024). This rule removes a letter preference that acts on guesses and leaves a marginal excess over the one-ability nulls, carried mostly by near-duplicate models (Sec. 3.3).

Spectral dimensionality measures have been applied to item-level benchmark matrices (Sha & Zhao, 2026), and letter preference and the pull of few-shot exemplars are known for single models (Zheng et al., 2024; Pezeshkpour & Hruschka, 2024; Zhao et al., 2021; Min et al., 2022; Ding, 2026) (App. A). We show that the across-model correctness matrix, which records no letter, still carries a letter preference, mostly one model's, as a direction.

2. A spectral test for cross-question structure

Entry c_{ki} of the correctness matrix $\mathbf{C}$ is 1 when row k answers question i correctly and 0 otherwise. Within one model, row k is pass k; across models, each row is one model. We keep the questions that vary. Within a model, these are the N_b *borderline* questions, whose pass rate f_i (the fraction of passes that answer the question correctly) satisfies $\varepsilon < f_i < 1 - \varepsilon$ with $\varepsilon = 0.05$ (App. C varies ε from 0.01 to 0.20). Across models, they are the *split* questions, which some models answer correctly and others do not. Let $\mathbf{R}$ be the Pearson correlation matrix of the kept columns over the rows. If the questions are independent and rows and columns grow in proportion, the eigenvalues of $\mathbf{R}$ follow the Marchenko–Pastur (MP) law, whose upper edge is $\lambda_+ = (1 + \sqrt{\gamma})^2$, with γ the ratio of columns to rows (Marčenko & Pastur, 1967). A direction gives each question a loading. A row's *score* on it is the projection of the row after each column is standardized.

Within one model. We compare the largest eigenvalue $\lambda_{\max}$ with λ_+. The Tracy–Widom statistic z measures how far $\lambda_{\max}$ lies above λ_+, in units of its typical fluctuation (Johnstone, 2001). We calibrate z against a null of 1000 matrices of independent Bernoulli entries at the observed pass rates, with the filter re-applied (App. C). The MP count is the number of eigenvalues above λ_+.

Across models. Each model answers every question in four *rotations*: rotation 0 is the original, and rotations 1, 2 and 3 move every option one, two or three letters later, cyclically. The *gold letter* of a question is the letter of its correct option. The reference is a fitted one-ability item response model (App. F). In the two-parameter logistic (2PL) model (Lord & Novick, 1968), model m answers question i correctly with probability $\sigma(a_i(\theta_m - b_i))$, where σ is the logistic function, θ_m the model's ability, and b_i and a_i the question's difficulty and slope. The Rasch model sets every slope equal (Rasch, 1960). The three-parameter (3PL) model adds a fixed guessing floor of 0.25. In the uniform know-or-guess (KG-u) null, a model knows a question, with the 2PL probability, in all four rotations at once. Otherwise it guesses uniformly among the four letters in each rotation. It is fitted to all four rotations and scored like the data.

With 0/1 answers, even one ability can lift a second eigenvalue above λ_+, through a second component tied to question difficulty: the *difficulty factor* of binary items (Ferguson, 1941; McDonald & Ahlawat, 1974). Instead of λ_+, we compare each eigenvalue λ_j with the jth eigenvalue of 1000 matrices simulated from the fitted null. Rank j *exceeds* the null when λ_j is above the null's 99th percentile. A null counts only if it is *valid*: the observed λ_1 lies between the 1st and 99th percentiles of its draws. λ_1 measures how tightly ability ties the answers together, so a null that misses it misjudges the eigenvalues after it (App. F). Because an excess can spread over several ranks, the *family-wise test* uses one statistic M: the largest of the percentile ranks of λ_2, λ_3 and λ_4 among 2000 new draws of each null (fewer in removals and simulations; App. F). Each draw gets its own M against the other draws, and p_M is the share of draws whose M is at least the observed one. A *sibling group* of near-duplicate models holds versions or fine-tunes of one base model at one size: Phi-3/3.5-mini, Llama-3/3.1-8B, Qwen2/2.5-7B, and Mistral-7B v0.2/v0.3 with Zephyr-7B. We fixed this test, the validity rule and the sibling groups before running the tests, but after computing a similar statistic and the model-pair correlations on the same data. Analyses that remove single models or model pairs are post hoc, as are the size, power and difficulty analyses of Sec. 3.3.

Runs. Within one model (App. B), an *independent-draw run* gives every (question, pass) pair its own seed, as self-consistency does. In a *shared-seed run*, all questions of a pass use one seed. In a *shared-prompt run*, each pass draws five MMLU development questions with their answer letters and places them as exemplars before every question of the pass. Ten independent-draw runs use temperature 1: five models on 500 MMLU test questions (Hendrycks et al., 2021) (ten from each of 50 subjects), three models on HellaSwag (Zellers et al., 2019), and Qwen2.5-7B on GSM8K (Cobbe et al., 2021) at two output lengths. Two more use Qwen2.5-7B on MMLU at temperatures 0.5 and 1.5. These twelve runs have $K = 30$ to 100 passes (Table A1). One shared-seed and two shared-prompt runs use MMLU with $K = 100$. Across models, 24 models of 0.5B to 22B parameters from ten families answer the 500 MMLU questions in all four rotations by greedy decoding (always the most probable answer). A reply that cannot be read as one answer (a letter, or a number for GSM8K) is unparseable and scored wrong: at most 1.65% of replies in the

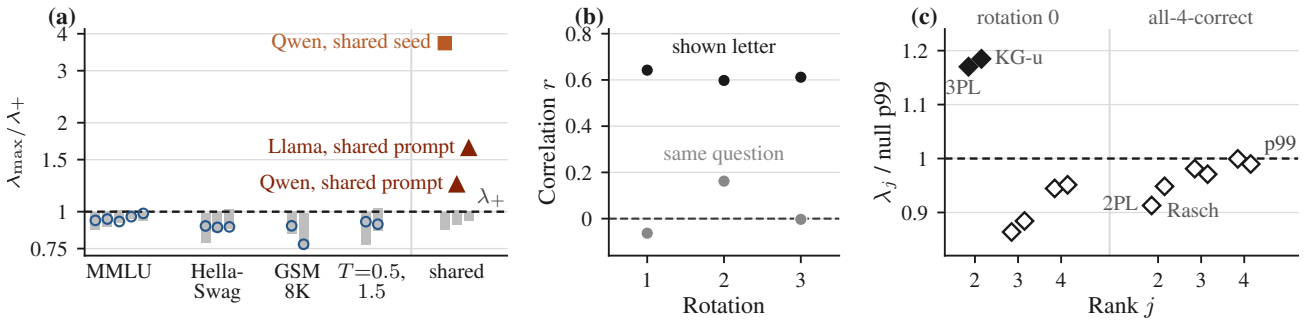


Figure 1. **(a)** Within one model: $\lambda_{\max}/\lambda_+$ (log axis) of each run; grey bar: 5–95% of its null; circles: independent draws; filled: an eigenvalue above λ_+. **(b)** Across models: correlation, over questions, of each question's eigenvector-2 loading in rotations 1–3 with the rotation-0 mean loading of the questions whose gold letter it now shows, and with its own rotation-0 loading. **(c)** λ_2–λ_4 of the 24-model matrix in rotation 0 and all-4-correct (Sec. 3.3), divided by the 99th percentile of two valid nulls in 1000 draws (rotation 0: 3PL left, KG-u right; all-4-correct: 2PL left, Rasch right); filled: above it.

within-model runs and at most 12.4% of one model's replies in rotation 0 (App. B).

3. Results

3.1. Within one model: structure only from a shared component

In all 12 independent-draw runs the largest eigenvalue lies below λ_+ (Fig. 1a). Every z falls between the 10th and 84th percentiles of its null, and the MP count stays zero at six borderline thresholds (App. C). To measure power, we plant a factor shared by the borderline questions of a pass, with latent correlation ρ between them. At these pass rates their 0/1 answers correlate at about $\rho/2$ (App. D). At $\rho = 0.05$ the MP count detects the factor in every simulated matrix at Llama-3-8B's $N_b = 173$, but in 47% at Qwen2.5-7B's $N_b = 43$. A cluster of nine questions at $\rho = 0.2$ is detected in at most 24%. The test can miss weak structure in a run with few borderline questions, or a small cluster.

In the shared-seed and shared-prompt runs, z exceeds every draw of its null and two to four eigenvalues exceed λ_+. We trace each run's top direction to its source (App. E). vLLM's sampler divides each token's probability by a random number and emits the token with the largest ratio (Gumbel-max sampling). With a shared seed, each letter token gets the same random number at a given generation step in every question of the pass, so a letter that draws a favorable number wins in many questions at once. In Qwen2.5-7B's shared-seed run, the gold letter explains $\eta^2 = 0.73$ of the variance of the top eigenvector's loadings (permutation $p = 0.0002$). With a shared prompt, Llama-3-8B copies the exemplars' answer letters, and its top direction groups by gold letter ($\eta^2 = 0.53$, $p = 0.0002$). Qwen2.5-7B's shared-prompt direction does not group by gold letter ($p = 0.094$), yet the exemplar set still couples its answers: two passes that drew a common exemplar answer more alike than two that did not ($p = 0.0005$). A shared seed leaves each question's answer distribution unchanged, so per-question self-consistency is unaffected, but the passes are no longer independent members of an ensemble.

3.2. Across models: the second direction is letter preference

In rotation 0 the four largest eigenvalues of the 24-model matrix (460 split questions, $\lambda_+ = 28.92$) are 85.9, 49.9, 29.7 and 27.3. Eigenvector 1 is the ability direction: the models' scores on it correlate with accuracy at $r = 0.94$. Only nulls that let a model guess right pass the validity rule: the observed λ_1 lies above all draws of the 2PL and Rasch nulls, but inside those of the 3PL and KG-u nulls. Of ranks 2 to 4, only rank 2 exceeds the 3PL and KG-u nulls (Fig. 1c). The family-wise test rejects both, and a valid null with letter guessing ($p_M = 0.002$ each; App. F).

The loadings of eigenvector 2 group by gold letter ($\eta^2 = 0.48$, $p = 0.0002$), and the models' scores on it correlate with their share of D answers at $r = -0.90$ and of A answers at $r = 0.76$. Within one rotation a question's gold letter is tied to its content, so this grouping cannot separate letter from content. Rotating the options separates them. In rotations 1–3 the scores on eigenvector 2 stay close to those of rotation 0 ($|r|$ from 0.95 to 0.98). We correlate each question's loading with two rotation-0 loadings: its own, and the mean loading of questions whose gold letter it now shows. A direction of question content would follow each question's own loading. Eigenvector 2 follows the shown letter instead ($r = 0.60$ to 0.64, against -0.06 to 0.16; Fig. 1b). An own-loading correlation of up to 0.16 leaves room for a small part that follows content. A *soft-preference* null combines one ability with a fitted letter preference that acts on every choice. It is valid here, and the family-wise test does

not reject it ($p_M = 0.079$), but only 2 of 200 datasets simulated from its fit give λ_2 as extreme a percentile ($p = 0.010$; App. F). The case for letter preference rests on the rotation test.

The direction comes mostly from one model, Llama-3.2-1B. A know-or-guess model fitted to all four rotations estimates that it picks A nine times in ten when it guesses. Removing Llama-3.2-1B lowers λ_2 from 49.9 to 35.0, below the 99th percentile of the 3PL null refitted to the other 23 models (40.1). Removing any other single model leaves λ_2 between 47.1 and 52.4. Without Llama-3.2-1B a weaker excess remains at rank 4, and the family-wise test still rejects the 3PL null ($p_M = 0.003$).

3.3. After rotation: a marginal excess, mostly from two model pairs

We count a model correct on a question only if it is correct in all four rotations. Over the rotations the gold option appears once at each letter, so on every question a model that guesses by letter alone is correct in all four with the same probability, $\prod_L \pi_m(L) \leq 1/256$, where $\pi_m(L)$ is its probability of guessing letter L. This *all-4-correct* score removes a letter preference that acts on guesses. We cannot test whether a preference that also acts on partly known choices survives, because the soft-preference null is invalid here (App. F). The all-4-correct matrix (391 split questions, $\lambda_+ = 25.36$) has eigenvalues 92.1, 29.0, 23.3 and 20.8.

Rank 2. With guessing nearly excluded, the validity pattern of rotation 0 reverses: the 2PL and Rasch nulls are valid, and the 3PL and KG-u nulls are not. In 2000 draws, the 2PL and Rasch 99th percentiles of λ_2, 31.9 and 30.2, lie above λ_+, and the observed λ_2 lies below both (Fig. 1c). In the 2PL and Rasch nulls, eigenvector 2 follows question difficulty: its loadings correlate with the share of models that answer each question correctly at a median $|r|$ of 0.62 and 0.63 (observed: 0.57; App. F). Could the test miss a second ability? We simulate from the 2PL fit and give the 125 split STEM questions a second ability, independent of the first and with the first ability's spread and median slope. Rank 2 then exceeds the null's 99th percentile in 69% (2PL) and 78% (Rasch) of datasets, and the family-wise test rejects in 91% and 93%. At half that strength the family-wise test rejects in about 20%, against 7% with no second ability, so a weaker second ability could be missed (App. F).

Ranks 3 and 4. In 2000 null draws, rank 3 lies at the 98.5th (2PL) and 97.0th (Rasch) percentile and rank 4 at the 99.1st and 98.6th. Rank 4 sits at the 2PL 99th percentile: 0.03 above it in these draws and 0.01 below it in the 1000 draws of Fig. 1c. The family-wise test, which pools ranks 2 to 4, rejects both nulls ($p_M = 0.026$ and 0.041; Table A5). The excess is marginal. In datasets simulated from the one-ability fits, the test, validity rule included, rejects in up to 8.7% at the 5% level. Calibrated against those datasets, the observed p_M are 0.017 to 0.057. Dropping the 217 of the 500 questions with any unparseable reply ends the rejection, but leaves only 236 split questions (App. B).

Source. Keeping one model of each sibling group removes five models and ends the rejection in all 24 ways of choosing. Removing five other models at random keeps the rejection in all 24 trials (App. F). From these results we chose two pairs post hoc, Phi-3-mini/Phi-3.5-mini and Llama-3-8B/Llama-3.1-8B. Removing one model from each raises p_M to 0.22–0.69 (2PL) and 0.32–0.38 (Rasch), above all 190 removals of two other models ($p_M \leq 0.133$). Removing Llama-3.2-1B alone also ends the rejection but raises p_M less (0.059 under the 2PL null and 0.089 under the Rasch null), so part of the excess still depends on the letter-biased model. Two simple baselines point to the same models. Letter shares flag Llama-3.2-1B among three letter-biased models, and correlations of 2PL residuals rank the two pairs first and second of 276 model pairs (App. F).

4. Discussion

The directions we find beyond ability come from a shared seed or prompt within a model and, across models, from a letter preference, mostly of one model. After rotation, the test detects only a marginal excess, carried mostly by two pairs of near-duplicate models and partly by Llama-3.2-1B. The test could miss a second ability half as strong as the first, so it does not show that the ensemble has no shared knowledge gaps. We did not test why the two pairs fail on the same questions: knowledge gaps from common training data, contamination or another shared cause. In the ensemble, architecture, family, scale, training data and instruction tuning vary together. Nor did we test whether item-level factor and item response analyses (Kipnis et al., 2025; Sha & Zhao, 2026; Yao et al., 2026) or shared wrong answers (Kim et al., 2025) carry the same letter confound.

We test correctness only, not log-probabilities, reasoning traces or an uncertainty score, and we did not simulate power for structure carried by a few models, so other structure may remain. The within-model runs cover only instruction-tuned 1.7B–8B models, the ensemble only 500 MMLU questions, and per-rank thresholds are uncorrected for the nulls and scores we tried (App. F).

Before an ensemble's errors, or a cross-model epistemic term such as that of Hamidieh et al. (2026), are read as epistemic, we recommend drawing every sample with its own seed, rotating the options, scoring a question correct only if it is correct in every rotation, testing the remaining spectrum against valid one-ability nulls, and checking whether near-duplicate models carry what remains.

References

Bai, Z. and Silverstein, J. W. *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics. Springer, New York, 2 edition, 2010. doi: 10.1007/978-1-4419-0661-8.

Bao, Z., Pan, G., and Zhou, W. Tracy–Widom law for the extreme eigenvalues of sample correlation matrices. *Electronic Journal of Probability*, 17:1–32, 2012. doi: 10.1214/EJP.v17-1962. arXiv:1110.5208.

Burnell, R., Hao, H., Conway, A. R. A., and Hernández-Orallo, J. Revealing the structure of language model capabilities. *arXiv preprint arXiv:2306.10062*, 2023.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Corvelo Benz, N. L., Tsirtsis, S., Straitouri, E., Chatzi, I., Artola Velasco, A., Thejaswi, S., and Gomez-Rodriguez, M. Evaluation of large language models via coupled token generation. In *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 300 of *PMLR*, pp. 2215–2223, 2026.

Ding, K. When LLMs agree, are they right? Auditing self-consistency and cross-model agreement as confidence signals. *arXiv preprint arXiv:2607.08065*, 2026.

Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024. doi: 10.1038/s41586-024-07421-0.

Fei, X., Zhang, Y., Carneiro, S. A., and Vazirgiannis, M. Every wrong answer counts: Option-level psychometrics for LLM multiple-choice benchmarks. *arXiv preprint arXiv:2608.02966*, 2026.

Ferguson, G. A. The factorial interpretation of test difficulty. *Psychometrika*, 6(5):323–329, 1941. doi: 10.1007/BF02288588.

Hamidieh, K., Thost, V., Gerych, W., Yurochkin, M., and Ghassemi, M. Complementing self-consistency with cross-model disagreement for uncertainty quantification. In *International Conference on Learning Representations*, 2026.

Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021.

Hüllermeier, E. and Waegeman, W. Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3):457–506, 2021. doi: 10.1007/s10994-021-05946-3.

Jiang, T. The limiting distributions of eigenvalues of sample correlation matrices. *Sankhyā*, 66(1):35–48, 2004.

Johnstone, I. M. On the distribution of the largest eigenvalue in principal components analysis. *Annals of Statistics*, 29(2): 295–327, 2001. doi: 10.1214/aos/1009210544.

Kendall, A. and Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Kim, E. M., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 30038–30066, 2025.

Kipnis, A., Voudouris, K., Schulze Buschoff, L. M., and Schulz, E. metabench – a sparse benchmark of reasoning and knowledge in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.

Lakshminarayanan, B., Pritzel, A., and Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Liu, Y., Duan, H., Zhang, Y., Li, B., Zhang, S., Zhao, W., Yuan, Y., Wang, J., He, C., Liu, Z., Chen, K., and Lin, D. MMBench: Is your multi-modal model an all-around player? In *Computer Vision – ECCV 2024*, volume 15064 of *Lecture Notes in Computer Science*, pp. 216–233. Springer, 2024. doi: 10.1007/978-3-031-72658-3_13.

Lord, F. M. and Novick, M. R. *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968. With contributions by Allan Birnbaum.

Maimon, A., Cohen, A. D. N., Vishne, G., Ravfogel, S., and Tsarfaty, R. From benchmarks to skills: Low-rank factors for LLM evaluation. *Transactions of the Association for Computational Linguistics*, 14:1639–1653, 2026. doi: 10.1162/tacl.a.745.

Manakul, P., Liusie, A., and Gales, M. J. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 9004–9017, 2023. doi: 10.18653/v1/2023.emnlp-main.557.

Marčenko, V. A. and Pastur, L. A. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967. doi: 10.1070/SM1967v001n04ABEH001994.

McDonald, R. P. and Ahlawat, K. S. Difficulty factors in binary data. *British Journal of Mathematical and Statistical Psychology*, 27(1):82–99, 1974. doi: 10.1111/j.2044-8317.1974.tb00530.x.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11048–11064, 2022. doi: 10.18653/v1/2022.emnlp-main.759.

Pezeshkpour, P. and Hruschka, E. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pp. 2006–2017, 2024. doi: 10.18653/v1/2024.findings-naacl.130.

Pillai, N. S. and Yin, J. Edge universality of correlation matrices. *Annals of Statistics*, 40(3):1737–1763, 2012. doi: 10.1214/12-AOS1022. arXiv:1112.2381.

Rasch, G. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danish Institute for Educational Research, Copenhagen, 1960.

Ravfogel, S., Svete, A., Snæbjarnarson, V., and Cotterell, R. Gumbel counterfactual generation from language models. In *International Conference on Learning Representations*, 2025.

Ruan, Y., Maddison, C. J., and Hashimoto, T. Observational scaling laws and the predictability of language model performance. In *Advances in Neural Information Processing Systems*, 2024.

Sha, T. and Zhao, S. BenchScope: How many independent signals does your benchmark provide? *arXiv preprint arXiv:2603.29357*, 2026.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Yao, L. H., Jarvis, N., Zhan, T., Ghosh, S., Liu, L., and Jiang, T. A geometric lens on LLM abilities through joint embedding item response theory. *Transactions on Machine Learning Research*, 2026. URL https://openreview.net/forum?id=VQe6p9wn5g. arXiv:2509.22888.

Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 4791–4800, 2019. doi: 10.18653/v1/P19-1472.

Zhao, Z., Wallace, E., Feng, S., Klein, D., and Singh, S. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, pp. 12697–12706, 2021.

Zheng, C., Zhou, H., Meng, F., Zhou, J., and Huang, M. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, 2024.

A. Background

Option position. Ding (2026) shuffle the options and find that part of a model's self-agreement depends on option position. In the option-level response model of Fei et al. (2026), a per-model positional preference barely helps to predict held-out correctness (accuracy 0.840 with it, 0.838 without). Such a preference can still create a second direction in a correctness matrix (Section 3.2).

Random-matrix tests for the number of factors. A factor shared by many variables lifts an eigenvalue of their sample correlation matrix above the MP edge λ_+ of Section 2, because n variables with pairwise correlation κ have a population eigenvalue of $1 + (n-1)\kappa$. This eigenvalue grows with n, so an eigenvalue test can detect a correlation smaller than the sampling noise of one pairwise correlation, about $1/\sqrt{K}$. The MP law holds for sample covariance matrices of independent entries with finite variance (Marčenko & Pastur, 1967; Bai & Silverstein, 2010), and for sample correlation matrices of i.i.d. entries (Jiang, 2004). For Gaussian data, Johnstone (2001) showed that the largest eigenvalue, minus the edge and divided by $\sigma_{\mathrm{TW}} = (1+\sqrt{\gamma})(1+1/\sqrt{\gamma})^{1/3}K^{-2/3}$, converges to the Tracy–Widom distribution. For sample correlation matrices, Bao et al. (2012) proved this limit for matrices formed without centering, with symmetrically distributed entries and fewer variables than observations. With centering, they proved it only for Gaussian entries. Pillai & Yin (2012) removed the symmetry condition and allowed more variables than observations, still without centering. Our $\mathbf{R}$ is centered at the sample means of Bernoulli columns with unequal pass rates. No theorem we know covers sample-mean centering of non-Gaussian entries, so we calibrate z with the Monte Carlo null instead.

Item response models. These models come from educational testing, where each test-taker has an ability and each question a difficulty. Here the test-takers are language models. Our 3PL form for four options is $0.25+0.75\,\sigma(a_i(\theta_m - b_i))$. Rasch, 2PL and 3PL are one-factor models: ability is the only source of dependence between questions.

B. Data collection

Questions. MMLU: the first 50 subjects in alphabetical order, ten questions drawn at random from each. HellaSwag: 200 validation items, sampled at random within each activity label. GSM8K: the first 100 test problems. The shared-prompt exemplars come from the 285 MMLU development questions. Pass k draws its five at random from all subjects, independently of the other passes, so two passes can share an exemplar. Every question of a pass sees the same five, and both shared-prompt runs use the same sets.

Prompts and decoding. Each model receives one user message in its own chat template. Phi-2, which has none, receives the raw text. Prompts are tokenized with each model's own tokenizer and checked against an independent reference tokenization. In shared-prompt runs, the message places the five exemplars, each followed by its answer letter, between the instruction and the question. Generation uses vLLM 0.23.0 in bfloat16 with temperature T, top-$p = 1$, top-k disabled, min-$p = 0$, repetition penalty 1.0, no presence or frequency penalty, and at most 8 new tokens for MMLU and HellaSwag and 256 (or 1024) for GSM8K. Greedy decoding sets $T = 0$. In independent-draw and shared-prompt runs, every sample (k, i) has a seed that no other sample of the run uses. In shared-seed runs, all questions of pass k share one seed, which changes from pass to pass. vLLM gives each seeded request its own random-number generator. A request's sampling noise then depends on its seed, not on its position in a batch, and requests with different seeds share no noise (App. E).

Rotations and replicates. The 24 models answer every question in all four rotations, with deterministic kernel settings. GPU kernels can still differ between engine starts, so we collected the ensemble twice, from separate engine starts (replicates A and B). The paper reports replicate A.

Scoring. A multiple-choice reply is read as letter X when it begins with X, optionally after "Answer:", a parenthesis or bold markers, and X is followed by punctuation, a line end, or words that do not negate or hedge it. It is also read as X when its first line says "the answer is X" or a similar phrase. A reply that names several letters, negates its letter, or fits neither rule is unparseable and scored wrong. The GSM8K prompt asks for the final answer after "####". A reply is correct when the number after its last "####", or else its last number, equals the gold answer.

Unparseable replies. In the within-model runs, at most 1.65% of replies are unparseable (Llama-3-8B on HellaSwag), and the shared-seed run and Qwen2.5-7B's shared-prompt run have none. In the ensemble, Mistral-7B v0.2 has 12.4% and every other model at most 9.6%. We either drop every question with an unparseable reply or replace each unparseable reply by the mean correctness of the question's parseable replies. Either way, every count of the independent-draw runs stays at zero, and the MP and buffered counts of Llama-3-8B's shared-prompt run stay at two. Across models, we rerun the valid nulls on both variants (Table A5). On all-4-correct, the family-wise test rejects after imputation and without the 95 questions whose score depends on an unparseable reply. Dropping all 217 questions with any unparseable reply ends the rejection (Section 3.3). In rotation 0, imputation leaves rank 2 above the 99th percentiles of the letter-aware (App. F) and 3PL nulls. Dropping the 112 questions with an unparseable rotation-0 reply moves rank 2 to the 98.3rd percentile of the letter-aware null. It stays above the 3PL null, and the family-wise test rejects under both ($p_M = 0.013$ and 0.004).

Truncation on GSM8K. At 256 new tokens, 55% of Qwen2.5-7B's chain-of-thought replies reach the limit. Most of them stop before the final answer, and the scorer reads the last number of an unfinished calculation: 13.6% of truncated replies are correct, against 96.7% of complete ones. Accuracy is 51.0% at 256 tokens and 90.3% at 1024. The 256-token run still tests independence: truncation changes which answers are correct but does not couple two questions of a pass.

C. Test statistics and checks

Assumption C.1 (Independent draws). Given the model and the prompts, the entries c_{ki} are Bernoulli variables with $\Pr(c_{ki} = 1) = p_i$, independent across questions i and passes k.

Table A1. The test on every within-model run ($\varepsilon = 0.05$). λ_+: MP edge; TW z: Tracy–Widom statistic. Counts: MP (above λ_+) / buffered (above $1.05\,\lambda_+$, a margin for the fluctuation of $\lambda_{\max}$). GSM8K runs use $T = 1$.

Run	N_b	λ_+	$\lambda_{\max}$	TW z	Counts
Independent, MMLU ($K = 100, T = 1$)					
Qwen2.5-7B	43	2.74	2.56	−1.72	0/0
Mistral-7B-v0.3	83	3.65	3.45	−1.79	0/0
Llama-3-8B	173	5.36	4.97	−3.05	0/0
SmolLM2-1.7B	406	9.09	8.74	−2.17	0/0
Phi-3-mini	271	7.00	6.91	−0.65	0/0
Independent, HellaSwag ($K = 50, T = 1$)					
Qwen2.5-7B	16	2.45	2.19	−1.60	0/0
Mistral-7B-v0.3	65	4.58	4.05	−2.71	0/0
Llama-3-8B	86	5.34	4.75	−2.90	0/0
Independent, Qwen2.5-7B ($K = 30$)					
GSM8K, 256 tok.	62	5.94	5.32	−2.06	0/0
GSM8K, 1024 tok.	19	3.22	2.50	−2.97	0/0
MMLU, $T = 0.5$	25	3.66	3.39	−1.08	0/0
MMLU, $T = 1.5$	85	7.20	6.53	−2.07	0/0
Shared component, MMLU ($K = 100, T = 1$)					
Qwen, seed	41	2.69	10.00	+70.2	4/4
Qwen, prompt	93	3.86	4.77	+7.9	2/2
Llama, prompt	264	6.89	11.31	+30.9	2/2

Under Assumption C.1 two borderline questions have population correlation zero, so the population correlation matrix of the borderline columns is the identity. The filter keeps or drops each question on its own column alone, so the kept columns remain independent. If every p_i stays a fixed distance from ε and $1 - \varepsilon$, the probability that f_i falls on the wrong side of a threshold decreases exponentially in K, and the kept set equals the set of questions with $\varepsilon < p_i < 1 - \varepsilon$ with probability tending to 1 as K and N grow in proportion.

No theorem we know gives the limit of z for $\mathbf{R}$ (App. A). At $K \leq 100$ the distribution of z is shifted from the Tracy–Widom law. Every z of the 12 independent-draw runs is negative, and their Monte Carlo nulls of z have means between -1.98 and -1.47. A z above every draw of its Monte Carlo null, as in the three shared-component runs, rejects Assumption C.1. The rejection does not say whether the dependence lies between questions or between passes.

Checks of independence. Three further checks on the independent-draw runs agree with Assumption C.1. First, a component that makes the questions of a pass right or wrong together would make the number of borderline questions a pass gets right vary more than independent draws at the pass rates predict. In the 12 runs the ratio of the observed to the predicted variance of this number lies between 0.61 and 1.09. No run is significantly overdispersed against a column-shuffle null (smallest $p = 0.27$). The two lowest

ratios have one-sided underdispersion $p = 0.039$ and 0.045, and neither is significant after a Holm correction over the 48 tests (12 runs $\times$ 4 tests: over- and underdispersion, and two trends over the pass index). This check cannot detect a coupling that cancels in the number correct, such as that of the shared seed (App. E). Second, in the $T = 1$ MMLU runs of Qwen2.5-7B and Llama-3-8B, borderline questions of the same subject are no more correlated than other pairs (mean same-subject correlation -0.018, $p = 0.80$, and 0.002, $p = 0.37$). Third, a pairwise test compares the largest Cramér's V and mutual information over all pairs of borderline questions with their permutation null. It finds no coupled pair in correctness in any of the five MMLU runs at $T = 1$, nor in the answer letters of Qwen2.5-7B (largest V 0.41, null 99th percentile 0.72).

Borderline threshold. The threshold ε decides which questions enter the test, and with them N_b, λ_+ and the power. We repeat the test at $\varepsilon = 0.01$, 0.02, 0.05, 0.10, 0.15 and 0.20, and re-apply the filter at each threshold to the data and to every matrix of the run's Monte Carlo null (1000 matrices). Over all 12 independent-draw runs and six thresholds, no eigenvalue exceeds its edge, and every observed z lies below the 95th percentile of its null. The three shared-component runs keep at least two eigenvalues above their edge at every threshold. The six tests of a run are nested, not independent: raising ε removes the questions whose pass rates lie closest to 0 or 1. At $\varepsilon = 0.20$ the three smallest sets have 6, 7 and 10 questions (GSM8K at 1024 tokens, HellaSwag Qwen2.5-7B, MMLU at $T = 0.5$).

D. Power of the within-model test

We plant a factor that makes some questions of a pass right or wrong together, and measure how often each count detects it. The simulated matrices use the pass rates of two independent-draw MMLU runs, Qwen2.5-7B ($N_b = 43$) and Llama-3-8B ($N_b = 173$). The factor loads on all of a run's borderline questions (a diffuse factor) or on 30 or 9 of them, chosen at random in each matrix.

Each entry comes from a hidden standard normal score: $c_{ki} = 1$ when the score falls below $\Phi^{-1}(f_i)$, with Φ the standard normal distribution function, so question i is correct with probability f_i. A loaded question has score $\sqrt{\rho}\,u_k + \sqrt{1-\rho}\,e_{ki}$. The factor u_k is one standard normal draw shared by the loaded questions of pass k, and e_{ki} is independent standard normal noise. The other questions have score e_{ki}. The score keeps unit variance, so each question keeps its pass rate, and the scores of two loaded questions correlate at ρ. Their 0/1 entries correlate at about half of ρ at these pass rates, averaged over pairs of borderline questions.

Each matrix has all 500 questions and $K = 100$ passes, and goes through the test of Section 2 with the borderline filter re-applied. A count of one or more detects the factor. The detection rate is the power when $\rho > 0$ and the false-positive rate when $\rho = 0$. Each scenario has 500 matrices (2000 at $\rho = 0$).

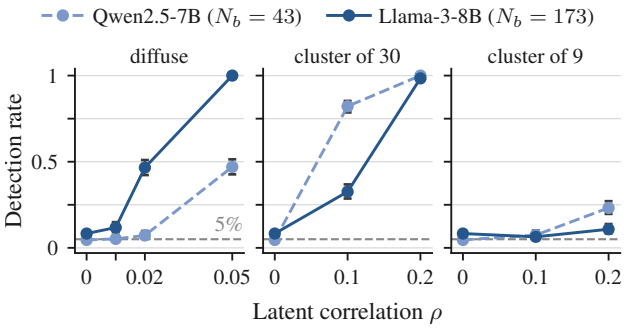


Figure A1. Detection rate of the MP count against the latent correlation ρ of a factor planted on all borderline questions or on 30 or 9 of them, at the pass rates of two MMLU runs. Whiskers: Wilson 95% intervals.

A diffuse factor is easier to detect at Llama-3-8B's 173 borderline questions, because it loads four times as many questions (Fig. A1). A fixed-size cluster is harder to detect there, because it is a smaller share of them. A 9-question cluster at $\rho = 0.2$ is detected in 23% of Qwen2.5-7B and 11% of Llama-3-8B simulations. The buffered count has fewer false positives and less power. At $\rho = 0$ the false-positive rates are 4.6% and 8.3% for the MP count and 0.65% and 0.35% for the buffered count. At $\rho = 0.05$ the buffered count detects a diffuse factor in 22% and 99.8% of simulations, against 47% and 100% for the MP count. We did not simulate other values of N_b.

E. What the shared component couples

Shared noise under a shared seed. At each generation step, a seeded request's random-number generator draws one exponential random number, the noise, for every token of the vocabulary. The request emits the token whose probability at temperature T, divided by its noise, is largest. Because the negative logarithm of an exponential random number is a Gumbel random number, this is Gumbel-max sampling, and each answer, taken alone, follows the model's distribution for its question. Two requests with the same seed draw the same noise at every step, whatever their prompts. Coupled generation and counterfactual sampling use this sharing on purpose (Corvelo Benz et al., 2026; Ravfogel et al., 2025). Suppose a shared-seed pass draws a small noise value for "B" at some step. The sampler divides the probability of "B" by that value in every question whose letter is generated at that step, and "B" gains in all.

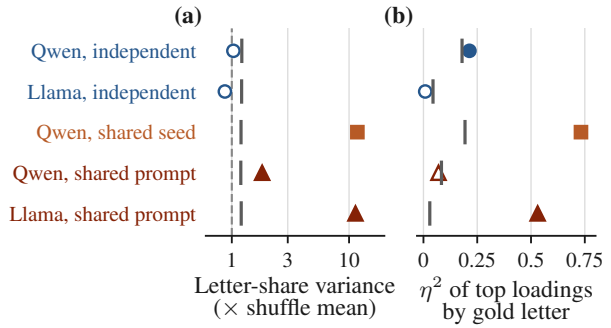


Figure A2. A shared seed or prompt makes a pass's letter shares vary beyond chance. Of the shared runs, only the seed run and Llama-3-8B's prompt run also group the top loadings by gold letter. **(a)** Across-pass variance of a pass's letter shares, summed over letters, divided by its mean when answers are shuffled across passes. **(b)** η^2 of the top eigenvector's loadings by gold letter. Grey tick: null 99th (a) or 95th (b) percentile; filled: beyond it.

What the directions follow. In Qwen2.5-7B's shared-seed run, the across-pass variance of a pass's letter shares (its fractions of A, B, C and D answers), summed over letters, is 11.8 times its mean when each question's answers are shuffled across passes (1.03 in its independent-draw run; Fig. A2a). The top eigenvector's loadings group by gold letter (Sec. 3.1). A pass whose noise favors one letter gains on questions with that gold letter and loses on the others. The gains and losses cancel in the number correct per pass (observed over predicted variance 0.83).

In its shared-prompt run, Llama-3-8B copies the exemplars' letters. Few-shot prompts are known to pull answers toward their labels (Zhao et al., 2021). Over passes, the number of a pass's five exemplars whose answer is letter L correlates with the pass's share of answers L at $r = 0.35$ to 0.47 for the four letters. The letter shares vary 11.3 times their shuffle mean, and the number correct per pass is overdispersed (ratio 1.51, $p = 0.001$). Neither the shared-seed direction nor this one groups by subject ($p = 0.23$ and 0.35).

In Qwen2.5-7B's shared-prompt run, the letter shares vary 1.8 times their shuffle mean but do not follow the exemplars' letters (r between -0.09 and 0.09), and the loadings do not group by gold letter ($\eta^2 = 0.07$). The exemplar set still couples its answers. Two passes that share an exemplar answer the borderline questions more alike than two that share none, by 10.7 standard deviations of a null that permutes exemplar sets across passes (5.9 for Llama-3-8B; permutation $p = 0.0005$ for both). On borderline questions, only two effects link this coupling to the exemplars' content. The loadings group weakly by subject ($p = 0.042$, uncorrected), and a same-subject exemplar raises accuracy on borderline questions by 3.2 points ($p = 0.033$).

Qwen2.5-7B's independent-draw run shows $\eta^2 = 0.21$ by gold letter ($p = 0.023$; Fig. A2b), although $\lambda_{\max}$ lies below λ_+. We do not interpret this value. The answer-letter test of App. C finds no pair of its borderline questions with letters associated beyond chance, but that test looks for single coupled pairs, not for a grouping of loadings by gold letter.

F. The 24-model ensemble

Roster. Five Qwen (Qwen2.5 at 0.5B, 1.5B, 3B and 7B, and Qwen2-7B), five Mistral (Mistral-7B v0.2 and v0.3, Zephyr-7B-beta, Ministral-8B, Mistral-Small-22B), four Llama (Llama-3.2 at 1B and 3B, Llama-3-8B, Llama-3.1-8B), three Phi (Phi-2, Phi-3-mini, Phi-3.5-mini), two Gemma-2 (2B and 9B), and one each of SmolLM2-1.7B, OLMo-2-7B, Yi-1.5-6B, DeepSeek-LLM-7B and Granite-3.1-8B: ten families, five with a single model. Phi-2 is a base model. The other 23 are instruction-tuned. Rotation-0 accuracy runs from 35.0% (Llama-3.2-1B) to 72.8% (Gemma-2-9B).

One-ability nulls. Every null is simulated conditional on every kept column staying split, so each of its draws has the observed number of columns and λ_+. Rasch is fitted by joint maximum likelihood. The fixed-margin null draws uniformly among 0/1 matrices with the observed row and column sums. The 2PL treats the abilities as fixed and integrates the question parameters over a normal population (marginal maximum likelihood). It is fitted to all 500 columns and simulated on the split ones. The Bock–Aitkin 2PL integrates the abilities instead, with slopes bounded by 4. The 3PL floor is 0.25 for every score. The letter-aware null is a know-or-guess model fitted to rotation 0, with each model's letter preference π_m estimated from all four rotations. The 4-rotation letter know-or-guess (KG-l) null is fitted like KG-u (Section 2) but guesses from π_m. Both 4-rotation nulls simulate every rotation.

Validity. A null counts only if the observed λ_1 lies between its 1st and 99th percentiles, for every score and null, the soft-preference null included. It excludes the 2PL, Rasch, fixed-margin, KG-l and Bock–Aitkin nulls in rotation 0, and the 3PL, fixed-margin, know-or-guess, Bock–Aitkin and SP nulls on all-4-correct (Table A2). We set the rule after the first analyses. The size simulation below tests it.

Construction family. All-4-correct is one of several ways to combine a model's four rotations. A letter preference that acts on choices the model partly knows can leak into any of them. Table A3 gives each score's directions beyond ability under its valid nulls. In rotation 0 without Llama-3.2-1B (Section 3.2), the letter-aware null also rejects ($p_M = 0.015$), and the letter grouping moves to eigenvector 3 ($\eta^2 = 0.21$). Ranks 3 and 4 lie at the 98.5th to 99.9th percentiles

Table A2. Percentile of the observed λ_1 among each one-ability null's draws; bold: valid (1 to 99). Draws per null: 1000 (rotation 0), 2000 (all-4-correct). –: fitted to rotation 0 only.

Null	Rotation 0	All-4-correct
2PL	100	**31.5**
Rasch	100	**54.2**
3PL	**76.4**	100
Letter-aware	**86.7**	–
Uniform know-or-guess	**98.7**	0.0
Soft preference	**85.7**	0.0

of the 3PL and letter-aware nulls. In their 1000 draws, rank 4 exceeds the 99th percentiles by 0.35 and 0.57. On ≥ 2 of 4 the family-wise test agrees with the per-rank rule ($p_M = 0.002$ under the 2PL, 0.86 under the 3PL). On ≥ 3 of 4 it rejects the Rasch, KG-l and 2PL nulls ($p_M = 0.008$, 0.007 and 0.009) but not KG-u (0.28). In its 2000 draws, the per-rank counts of the Rasch and KG-l nulls are 1 and 3, not the table's 2, because these ranks lie close to their 99th percentiles. On ≥ 1 of 4 only the SP null is valid, and the family-wise test also finds nothing ($p_M = 0.996$). The mean has no valid null. Removing the 23 questions whose options refer to other options ("all of the above", "A and B only") leaves the counts of rotation 0 and all-4-correct unchanged, and lowers those of ≥ 2 and ≥ 3 by one or two.

Soft-preference null. In the SP null, model m chooses option j of question i in a rotation with probability proportional to $\exp\big(b_{ij} + \mathbb{1}[j = g_i](a_i\theta_m + e_{mi})\big)\,\pi_m(L_j)$, where g_i is the gold option, b_{ij} is a fitted intercept of option j with $b_{ig_i} = 0$, L_j is the letter at which the rotation shows option j, and $e_{mi} \sim N(0, \tau^2)$ is shared by the four rotations of a cell and independent across models. Ability θ_m is the only link between questions. The letter preference π_m acts on every choice, even when the model partly knows the answer. The fit is penalized maximum likelihood over the parsed choices of all four rotations ($\tau = 1.17$). Simulations mask the cells unparseable in the data, and the rotation-0 result depends on this mask: the observed λ_2 lies at the 99.8th percentile of the null without it and at the 95.1st with it. Refitting inflates the slopes, so we calibrate: we simulate 200 datasets from the fit and refit each with its own null of 1000 draws. The calibrated p of rank j is the fraction of these datasets whose λ_j percentile is at least the observed one. In rotation 0 the SP null is valid (Table A2), and the calibrated p of λ_2 is 0.010 (Section 3.2), or 0.015 by the Monte Carlo convention $(1 + k)/(1 + n)$ with k of the $n = 200$ datasets at least as extreme. Without Llama-3.2-1B the null is invalid. On ≥ 3 of 4 and all-4-correct it is invalid (λ_1 at the 0th percentile), so its rejections there are not evidence.

Nulls with a letter preference. In rotation 0, rank 2 also exceeds the valid letter-aware null, and no other rank does. The family-wise test rejects this null ($p_M = 0.002$). The letter-aware null lets the preference act only when a model guesses. The soft-preference null above also lets the preference tilt choices that a model partly knows. In 200 datasets simulated from it with no second factor, rank 2 exceeds the letter-aware null in 60% and the 3PL null in 99.5% (each null refitted, 200 draws). Under such a preference, rank 2 often exceeds these nulls without a second factor.

Predictive check. The soft-preference model fails its predictive check, set before the run, against 500 datasets simulated from the fit. Per-model accuracy and letter shares fit. The number of rotations, 0 to 4, in which a model answers a question correctly does not: the simulations cover it in 82.5% of model-by-count cells, below the required 90%, and the observed number of split columns on ≥ 2 of 4 lies above all simulations. The weakest, letter-biased models have too many questions correct in exactly one rotation, as expected if they choose one letter almost deterministically. Most models, 21 of 24, also have too many correct in all four (largest excess 0.06). Pooled, the observed fractions are 0.196 for one rotation and 0.382 for four, against 0.177 and 0.366 simulated. The calibration removes the bias of the fitting pipeline only if each refit inflates the slopes as much as the first fit did, and it does not correct this misfit, so the soft-preference verdicts are not fully validated.

Family-wise test. The test uses 2000 new null draws, with seeds disjoint from all earlier runs. Its statistic M is the largest of the percentiles of λ_2, λ_3 and λ_4 among these draws, with ties counted as half. Each draw's own M_d is computed against the other draws, and $p_M = (1 + \#\{d : M_d \geq M\})/2001$, at level 0.05 (Table A5). Three null draws reach $M = 100$, so p_M cannot fall below 0.002, the value in rotation 0 and on ≥ 2 of 4. On 200 datasets simulated from the SP fit with no second factor (500 null draws each), the refitted 2PL null is valid on ≥ 2 and ≥ 3 of 4 in 98.5% and 100% of datasets, and valid and rejecting in 98.5% and 58% of all datasets. The refitted Rasch null is valid in 0% and 18%, and valid and rejecting in 0% and 11% of all datasets. The SP fit is valid on ≥ 2 but not on ≥ 3, so the ≥ 3 rates are only indicative. No simulation covers the KG-l test. We therefore draw no conclusion from the ≥ 2 2PL rejection or from the ≥ 3 rejections.

Size. We simulate 300 all-4-correct datasets from each fitted one-ability model, the 2PL and Rasch, with the nulls' own simulator. To each dataset we refit both nulls (500 draws each) and apply the validity rule and the family-wise test. Table A5 gives the rejection rates and the observed p_M calibrated against these datasets. The 2PL test rejects in 6.7% of datasets under 2PL truth and 8.7% under Rasch truth (95% intervals 4.1–10.1% and 5.7–12.4%), so it is not size-controlled by the bound of 7.5% we set before the

Table A3. Construction family and soft-preference (SP) null. Each score combines a model's four rotations: correct in rotation 0, in at least k of 4, or the mean. Valid nulls, each with its number of directions beyond ability by the per-rank rule: ranks $2, 3, \ldots$ that each exceed the null's 99th percentile, counted until one does not. Results for all 24 models and without Llama-3.2-1B (w/o); 1000 draws per null. "none": no valid null other than SP. LA: letter-aware; KG-l, KG-u: 4-rotation letter and uniform know-or-guess; BA: Bock–Aitkin 2PL. SP: rank-2 calibrated p; "inv.": invalid.

Score	Cols	λ_1–λ_4	Valid nulls: ranks beyond ability		SP: rank-2 cal. p
			24 models	w/o Llama-3.2-1B	24 / w/o
Rotation 0	460	85.9, 49.9, 29.7, 27.3	3PL 1, KG-u 1, LA 1	3PL, LA 0	0.010 / inv.
≥ 1 of 4	334	34.2, 27.2, 23.7, 21.4	none	BA 0	0.750 / inv.
≥ 2 of 4	444	108.3, 42.3, 30.2, 23.0	2PL 3, 3PL 0	2PL 1, 3PL 0	0.245 / 0.280
≥ 3 of 4	439	103.4, 37.8, 25.5, 22.1	Rasch 2, KG-l 2; 2PL, KG-u 0	2PL, KG-l, KG-u 0	inv. / 0.285
All 4	391	92.1, 29.0, 23.3, 20.8	Rasch, 2PL 0	Rasch, 2PL, KG-u 0	inv. / inv.
Mean	489	127.5, 43.1, 28.5, 25.7	none	none	inv. / inv.

run. Under Rasch truth the Rasch test rarely rejects (0.7%), mainly because the refitted Rasch null is valid in only 23% of those datasets. The validity rule is not sufficient. It misses wrong slopes, since the Rasch null is valid in every dataset simulated from the 2PL fit. It also declares the correct Rasch model, refitted by joint maximum likelihood, invalid in most datasets simulated from it.

Sibling models. We chose the sibling groups (Section 2) from model metadata. Models of one family at different sizes are not siblings. Keeping one model per group gives $2 \cdot 2 \cdot 3 \cdot 2 = 24$ subsets of 19 models. We refit both nulls to each subset, with 1000 draws, and none of the 24 rejects (Table A5). As a control, we remove at random five of the 15 models outside the groups, 24 times, again leaving 19 models. All 24 random subsets reject. The subsets of each kind overlap heavily, so the comparison is one contrast, not 24 replicates. The random removals include models that share a developer with the groups (Phi-2, Qwen2.5 at 0.5B to 3B, Llama-3.2, Ministral-8B and Mistral-Small), and are not matched in size to the group models (3.8B Phi, 7B to 8B for the others). On all-4-correct both nulls are valid in every sibling and random subset. In rotation 0 the letter direction does not depend on the siblings: the 3PL and letter-aware nulls are valid and reject in all 24 sibling subsets, while the 2PL and Rasch nulls are valid in none and the uniform know-or-guess null in 17.

Pairs and simple baselines. We remove one model of each of the two pairs of Section 3.3, in all four ways, and compare these four subsets with the 190 reference subsets that remove two of the other 20 models (1000 draws per subset; every null is valid in every subset). Under both nulls, each pair removal gives a larger p_M than every reference removal (Table A5), and 90% (2PL) and 74% (Rasch) of the reference removals still reject. Reference removals that include Qwen2-7B or Qwen2.5-7B reject more strongly (mean p_M 0.010 and 0.006 under the 2PL), so the Qwen pair is not part of the excess. The effect is specific to the two pairs. The Mistral near-duplicates do not carry it: removing v0.2 and v0.3 (residual rank 3 in Table A4) keeps the rejection ($p_M = 0.019$ under both nulls), as do the other two removals within the Mistral group (p_M 0.004 to 0.023). Nor does accuracy explain it. The pair models have rotation-0 accuracy 0.644 to 0.684, and no reference removal has both models inside that range. The 15 reference removals whose two models lie in the wider range 0.62 to 0.71 give p_M at most 0.052 (2PL) and 0.113 (Rasch), below all pair removals.

The letter-biased model contributes too. Without Llama-3.2-1B alone (Section 3.3), the uniform know-or-guess null becomes valid and gives $p_M = 0.15$. A second statistic, fixed with the test but not used for decisions, compares $\lambda_2 + \lambda_3 + \lambda_4$ with its null draws. Without Llama-3.2-1B it still gives $p = 0.023$ (2PL) and 0.016 (Rasch). Of the 19 reference removals that include Llama-3.2-1B, 12 (2PL) and 6 (Rasch) still reject, with p_M at most 0.099 and 0.130. Removing Llama-3.2-1B together with one model of each pair gives p_M 0.30 to 0.59 (2PL) and 0.38 to 0.60 (Rasch), close to the pair removals.

Table A4 gives two simple baselines. Rotation-0 letter shares flag the three models whose most frequent letter exceeds half of their replies. These shares count unparseable replies in the denominator, whereas the preference of 0.90 for A in Section 3.2 is estimated over guesses. Correlations of model pairs' residuals (observed minus 2PL-expected correctness over the 391 all-4-correct columns) rank the Phi and Llama pairs first and second of 276. The spectrum adds a test of whether such candidates exceed one ability.

Power. To measure what the all-4-correct verdict can exclude, we simulate from the valid all-4-correct 2PL fit plus a second ability. As in the 2PL null, each of the 391 kept columns draws (a_i, b_i) from its posterior. On the 125 of them from the 18 STEM subjects (mathematics, statistics, the natural sciences, computer science and electrical engineering), the logit gains $s\,\bar{a}\,\mathrm{sd}(\theta)\,z_m$. Here $z_m \sim N(0, 1)$

Table A4. Simple baselines. Top: rotation-0 letter shares (unparseable replies in the denominator) of the three models whose largest share exceeds 0.5; no other model's largest share exceeds 0.35. Bottom: correlation of model pairs' residuals after the all-4-correct 2PL fit, with rank among 276 pairs (mean -0.04, standard deviation 0.08).

Model	A	B	C	D
Llama-3.2-1B	0.81	0.04	0.09	0.05
Qwen2.5-0.5B	0.23	0.06	0.12	0.58
SmolLM2-1.7B	0.07	0.20	0.17	0.56

Model pair	r	**Rank**
Phi-3-mini, Phi-3.5-mini	0.40	1
Llama-3-8B, Llama-3.1-8B	0.27	2
Mistral-7B v0.2, v0.3	0.20	3
Mistral-7B v0.3, Llama-3-8B	0.19	4
Mistral-7B v0.2, Zephyr-7B	0.17	5
Qwen2-7B, Qwen2.5-7B	0.03	37

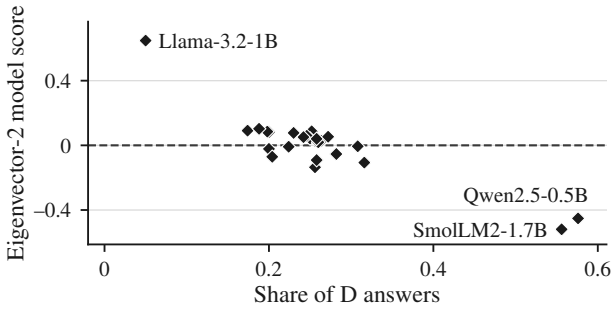


Figure A3. Each model's rotation-0 score on eigenvector 2 of the 24-model matrix against its share of D answers.

is independent of θ_m, $\bar{a} = 1.71$ is the median slope and $\mathrm{sd}(\theta) = 1$, so at $s = 1$ the second ability matches the first in spread and median slope. We refit both nulls to each of 200 datasets per s (500 draws each; rates in Table A5). At $s = 2$ the family-wise test and the rank-2 rule detect the second ability in at least 97% of datasets. The median λ_2 is 28.2 at $s = 0$, 28.3 at $s = 0.5$ and 33.8 at $s = 1$, against the observed 29.0. In the all-4-correct nulls the eigenvector-2 loadings follow question difficulty (Section 3.3). The observed $|r|$, 0.57, is just below the 2PL null's 5th percentile and inside the Rasch null's 5–95% range; in the rotation-0 3PL null the median $|r|$ is only 0.09 (observed 0.04).

Letter preference. In Figure A3, score and D share correlate at $r = -0.90$ (permutation $p = 10^{-4}$). Eigenvector 1 also carries a letter part ($\eta^2 = 0.22$ by gold letter), and eigenvectors 3 and 4 group by it more weakly but beyond chance ($\eta^2 = 0.087$ and 0.061, $p = 0.0002$ each). With gold letters nearly balanced (118 to 129 per letter), a letter preference leaves expected benchmark scores nearly unchanged.

Table A5. Family-wise test on all-4-correct (reject at $p_M < 0.05$). Subsets: number that reject, and the range or maximum of p_M. Pair removals: one Phi and one Llama model. Other two removed: the 190 removals of two other models (fraction that reject, maximum p_M). Unparseable: replies imputed, or questions dropped, either the 95 whose score they affect or all 217 that contain one (1000 draws). Size: rejection rate in 300 datasets simulated from each fitted one-ability model. Calibrated p_M: under 2PL and Rasch truth. Power: family-wise / rank-2 rule, 200 datasets per strength s (2PL fit).

	2PL	**Rasch**
Pct. of $\lambda_2, \lambda_3, \lambda_4$	71.8, 98.5, 99.1	94.6, 97.0, 98.6
p_M, 24 models	0.026	0.041
Sibling subsets	0/24 (0.09–0.65)	0/24 (0.12–0.69)
Random subsets	24/24 ($\leq$0.047)	24/24 ($\leq$0.046)
Pair removals	0/4 (0.22–0.69)	0/4 (0.32–0.38)
Other two removed	0.90 ($\leq$0.099)	0.74 ($\leq$0.133)
Unparseable: impute	0.019	0.042
drop 95	0.036	0.043
drop 217	0.23	0.12
Size, 2PL truth	0.067	0.067
Size, Rasch truth	0.087	0.007
Calibrated p_M	0.033, 0.047	0.057, 0.017
Power, $s = 0.5$	0.20 / 0.01	0.21 / 0.07
Power, $s = 1$	0.91 / 0.69	0.93 / 0.78

The remaining eigenvectors. The question loadings of the all-4-correct eigenvector 2 group by subject ($\eta^2 = 0.21$, $p = 0.001$). A cubic in question difficulty explains 39% of their variance, and without that trend the grouping is at chance ($p = 0.24$). Of the other seven directions tested (Table A3), only ranks 2 and 3 of ≥ 2 of 4 group by subject after this removal ($p = 0.010$ and 0.034, uncorrected over eight eigenvectors). On all-4-correct, the models' scores on eigenvector 4, below λ_+, group by family ($\eta^2 = 0.74$, $p = 0.006$). In rotation 0, only eigenvector 4 groups by family after a Holm correction ($p = 0.017$), and same-family model pairs correlate little more than other pairs (mean r 0.414 against 0.375, permutation $p = 0.085$).

Replicates and near ties. Replicate B differs from A in 6 of 48,000 chosen options, so it tests GPU nondeterminism only. Their eigenvalues differ by at most 0.081 over all matrices. In replicate B, λ_4 lies above the 2PL rank-4 99th percentile of 1000 draws (20.82 against 20.78; Monte Carlo $p = 0.009$), unlike replicate A. A near tie is a cell whose two most probable first-token letters differ by under 0.13 nats in log probability in either replicate (5.0% of cells). Replacing near ties by the majority answer over rotations changes no verdict. Without the 297 questions with a rotation-0 near tie (203 remain), rank 2 exceeds the 3PL but not the letter-aware null (both valid).